Statistics Notes

Weiguang Liu

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Chapter 1

Stochastic Convergence

From H. White's book on asymptotics.

Big O, small o. For $(b_n) \subset \mathbb{R}$, $b_n = O(c_n)$ if $c_n^{-1} b_n$ is eventually bounded. $b_n = o(c_n)$ if $c_n^{-1} b_n \longrightarrow 0$ as $n \longrightarrow \infty$.

Big O_p and small o_p. Let $(b_n) \subset \mathbb{R}$, then we say

1. $b_n = O_p(c_n)$ if for any $\varepsilon > 0$, there exists M_ε and N_ε such that for all $n > N_\varepsilon$,

$$P\left(\omega : \left|c_n^{-1} b_n\right| > M_\varepsilon\right) < \varepsilon \tag{1.1}$$

2. $b_n = o_p(c_n)$ if $c_n^{-1} b_n \xrightarrow{p} 0$.

Remark. If $b_n \in \mathbb{R}^k$ or $\mathbb{M}_{m \times n}$, then we mean convergence elementwise. Similarly for *a.s.* and convergence in probability.

1.1 Proposition (Algebra of O Notation). *The following results hold for O_p and o_p as well:*

1. $O(a_n) + O(b_n) = O(\max[a_n, b_n])$, $O(a_n)O(b_n) = O(a_n b_n)$;
2. $o(a_n) + o(b_n) = o(\max[a_n, b_n])$, $o(a_n)o(b_n) = o(a_n b_n)$;
3. $O(a_n) + o(b_n) = O(\max[a_n, b_n])$, $O(a_n)o(b_n) = o(a_n b_n)$;
4. *If a matrix $A_n = o(1)$ and a conforming vector $b_n = O(1)$ then $A_n b_n = o(1)$.*

1.2 Proposition (Uniform Continuous Mapping Theorem). *Let $b_n(\cdot) - c_n \longrightarrow 0$ a.s. where c_n is a nonrandom vector that doesn't necessarily converge. Let $g(\cdot) : \mathbb{R}^k \longrightarrow \mathbb{R}^k$ be a unifrom continuous function, then*

$$g(b_n(\cdot)) - g(c_n) \longrightarrow 0 \quad a.s. \tag{1.2}$$

Proof. For $\varepsilon > 0$, fix any ω such that $b_n(\omega) - c_n \longrightarrow 0$, by uniform continuity of g there exists $\delta(\varepsilon)$ such that for all x, y with $\|x - y\| < \delta$, $\|g(x) - g(y)\| < \varepsilon$. Eventually, $\|b_n - c_n\| < \delta$ for almost everywhere ω. □

1.3 Proposition (Matrix Inverse Is Uniformly Continuous). *For A_n uniformly positive definite, $(A_n)^{-1}$ is a uniformly continuous function.*

Proof. Similar to scalar inverse if you assume $b_n > \eta$. □

Chapter 2

Statistical Decision Theory

2.1 Statistical Set Up

First let's set up our playground.

Sample Space. Let $(\Omega, \mathcal{A}, \mu)$ be a probability space, call it the *experiment space*.

The *data* or a *sample* $X = (X_i)_{i \in \mathcal{J}}$ is a set of measurable functions $X_i : \Omega \longrightarrow \mathcal{X}$ where $\mathcal{X}$ is a *sample space* with σ-algebra $\mathcal{F}$.

Let $P_0 = P_X := X(\mu)$ be the (partially) unknown distribution of X. Usually we assume P_0 is in a family of probability measure on $(X, \mathcal{F})$

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}$$

Statistics. Statistics is about infering information about P_0 from observing X. For a functional T_0 on $\mathcal{P}$, we are interested in $T_0(P_0)$.

A *statistic* is a measurable function T:

$$T : \mathcal{X} \longrightarrow \mathbb{R}^k$$

Example (Measurement Error)**.** Let $X_i \in L^1(\Omega, \mathbb{R})$ for $i = 1, 2, \ldots, n$, where n is sample size. $\mathcal{X} = \mathbb{R}^n$. Measurability of $X = (X_i)_1^n$ is guaranteed if we take product measure on $\mathcal{X}$. If X_i are independent and identically distributed with P_0 on $\mathbb{R}$, then $P_{0,n} = \prod_{i=1}^n P_0$

on $(\mathbb{R}^n, \otimes\mathcal{B})$. The family of distribution we consider would be

$$\mathcal{P} = \left\{ \prod_i^n P : \quad \text{for some} \quad P \quad \text{on} \quad \mathbb{R} \right\} = \mu_0$$

Suppose we want to know the **mean** of P_0, then we take $T_0 : \mathcal{P} \longrightarrow \mathbb{R}$

$$T_0\left(\prod_i P\right) = \int_{\mathbb{R}} x \, \mathrm{d}P(x) \quad \text{and} \quad \mu_0 = T_0(P_{0,n})$$

A natural estimator(statistic) is $T : \mathcal{X} \longrightarrow \mathbb{R}$,

$$T(X) = \frac{1}{n} \sum_i^n X_i$$

Then we can say things about T. First it's **unbiased**:

$$E(T(X)) = P_n\left(\frac{1}{n} \sum_i^n X_i\right) = \frac{1}{n} P \otimes P \otimes \cdots \otimes P\left(\sum X_i\right) \quad \text{for all} \quad P \in \mathcal{P}$$

Asymptotics is more delicate. For example we can prove that $T(X)$ is consistent, here we define a sequence of statistics that depends on the sample size n,

$$T_n : \mathcal{X}_n = \mathbb{R}^n \longrightarrow \mathbb{R}$$

Consistency of $T_n(X)$ means that $T_n \xrightarrow{p} \mu_0$, which is guaranteed by the law of large numbers. But here we need to go back to the original experiment space.

Unbiasedness. A statistic T for T_0 is **unbiased** if for any $P \in \mathcal{P}$,

$$P(T(X)) = T_0(P)$$

Consistency of a Statistics. Let T_n be a sequence of statistics of an experiment, that is if we have a triangular array of data $X_{j,n}, 1 \leq j \leq n, n = 1, 2, \ldots$, $T_n : \mathcal{X}_n \longrightarrow (\mathcal{T}, d)$ a metric space, then we say T_n is a consistent estimator for

2.2 Risk

2.3 Group Families

See TPE. Also see three types of methds: explatory data analysis, decision theory and Bayesian.

2.4 Unbiasedness and Invariance

Chapter 3

Estimation

3.1 Unbiased Estimation

3.2 Bayes Estimators

3.3 Parametric Estimation

3.4 Nonparametric Estimation

Chapter 4

Hypothesis Testing in Parametric Models

4.1 UMP Tests

4.1 Neyman-Pearson Lemma. *Let $H_0 = (P_0)$, $H_1 = (P_1)$, let f_j be the p.d.f wrt a σ-finite measure ν, then*

1. *For every α, there exists a UMP test of size α, which is equal to*

$$T_*(X) = \begin{cases} 1 & f_1(X) > cf_0(X) \\ \gamma & f_1 = cf_0(X) \\ 0 & f_1(X) < cf_0(X) \end{cases}$$

   where $\gamma \in [0, 1]$ and $c \geq 0$.

2. *If T_{**} is a UMP test of size α, then*

$$T_{**}(X) = \begin{cases} 1 & f_1(X) > cf_0(X) \\ 0 & f_1(X) < cf_0(X) \end{cases} \quad a.s.\mathcal{P}$$

$\diamond$

4.2 Likelihood Ratio

Let $\mathcal{P} = \left\{ P_\theta \,:\, \theta \in \Theta \subset \mathbb{R}^k \right\}$ with pdf f_θ.

For a partition of Θ, we test

$$H_0 \,:\, \theta \in \Theta_0 \quad \text{against} \quad H_1 \,:\, \theta \in \Theta_1.$$

Likelihood Ratio. Let $l(\theta) = f_\theta(X)$, a **likelihood ratio** test is any test that rejects H_0 iff $\lambda(X) < c$ where $c \in [0, 1]$ and

$$\lambda(X) = \frac{\sup_{\Theta_0} l(\theta)}{\sup_{\Theta} l(\theta)}$$

Given α, we want to find c_α such that

$$\sup_{\theta \in \Theta_0} P_\theta(\lambda(X) < c_\alpha) = \alpha$$

4.3 Wald and Score Tests

4.4 Chi-squared and Goodness of Fit

4.5 Bayes Tests

Chapter 5

Hypothesis Testing in Nonparametric Models

5.1 Sign, Order and Rank Tests

5.2 Kolmogorov-Smirnov Test

5.3 Cranmer-von-Mises Test

5.4 Empirical Likelihood Ratio Test

Chapter 6

Confidence Sets

6.1 Confidence Sets

6.2 Asymptotic Confidence Sets

6.3 Bootstrap Confidence Sets

6.4 Simultaneous Confidence Sets